

Particle in a One-Dimensional Box

Energy Quantisation, Probability Density and Heisenberg Uncertainty

BPhO Computational Challenge 2026 — Task 7

Abstract

This report implements the exact stationary states of a non-relativistic particle confined by an infinite one-dimensional potential well. The energy spectrum and Born probability densities satisfy the official Task 7 requirements. The extension evaluates the position and momentum moments and shows analytically and computationally that every state obeys $\Delta x \Delta p \geq \hbar/2$. A finite-difference eigensolver provides an independent numerical check of the analytical solution.

1 Physical model

The potential is zero inside a box of width a and infinite outside:

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & x \leq 0 \text{ or } x \geq a. \end{cases} \quad (1)$$

The wavefunction must therefore vanish at both walls, $\psi(0) = \psi(a) = 0$. Within the box the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi. \quad (2)$$

The boundary conditions select positive integer quantum numbers $n = 1, 2, \dots$ and normalized eigenstates

$$\psi_n(x, t) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-iE_n t/\hbar}. \quad (3)$$

The corresponding energy spectrum is

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2} = \frac{n^2 h^2}{8ma^2}. \quad (4)$$

The computational baseline uses an electron and $a = 1.00 \text{ nm}$.

2 Probability density

The Born probability density is

$$|\psi_n(x, t)|^2 = \frac{2}{a} \sin^2\left(\frac{n\pi x}{a}\right). \quad (5)$$

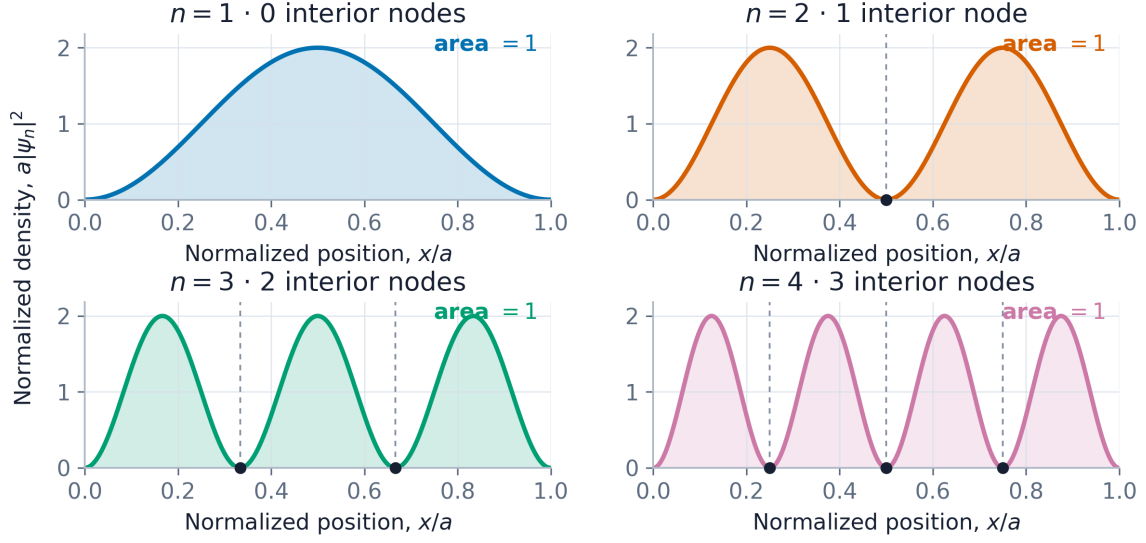
The factor $e^{-iE_n t/\hbar}$ has unit modulus, so a single energy eigenstate has a time-independent probability density. Normalisation follows from

$$\int_0^a |\psi_n|^2 dx = \frac{2}{a} \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = 1. \quad (6)$$

State n has $n - 1$ interior nodes at $x = ka/n$, where $k = 1, \dots, n - 1$.

First four stationary-state probability densities

$a|\psi_n|^2 = 2\sin^2(n\pi x/a)$ · each curve integrates to one and vanishes at both walls



Higher-energy states have more nodes, but every density remains symmetric about the centre of the box.

Figure 1: Dimensionless probability densities for the first four stationary states. The area under every curve is one.

3 Position uncertainty

The expectation value of position is

$$\langle x \rangle = \int_0^a x |\psi_n|^2 dx = \frac{a}{2}. \quad (7)$$

The second moment is

$$\langle x^2 \rangle = \frac{2}{a} \int_0^a x^2 \sin^2\left(\frac{n\pi x}{a}\right) dx \quad (8)$$

$$= a^2 \left(\frac{1}{3} - \frac{1}{2n^2\pi^2} \right). \quad (9)$$

Consequently,

$$(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2 \quad (10)$$

$$= a^2 \left(\frac{1}{12} - \frac{1}{2n^2\pi^2} \right), \quad (11)$$

and

$$\Delta x = a \sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}}. \quad (12)$$

4 Momentum uncertainty

The momentum operator is

$$\hat{p} = -i\hbar \frac{d}{dx}. \quad (13)$$

Because each spatial eigenfunction is real and vanishes at the boundaries,

$$\langle p \rangle = \int_0^a \psi_n^* \left(-i\hbar \frac{d}{dx} \right) \psi_n dx \quad (14)$$

$$= -\frac{i\hbar}{2} [\psi_n^2]_0^a = 0. \quad (15)$$

Applying the momentum operator twice gives

$$\langle p^2 \rangle = \int_0^a \psi_n^* \left(-\hbar^2 \frac{d^2}{dx^2} \right) \psi_n dx \quad (16)$$

$$= \frac{n^2 \pi^2 \hbar^2}{a^2}. \quad (17)$$

Thus

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \frac{n\pi\hbar}{a}. \quad (18)$$

5 Verification of the uncertainty principle

Multiplying Eqs. (12) and (18) gives

$$\Delta x \Delta p = \hbar \sqrt{\frac{n^2 \pi^2}{12} - \frac{1}{2}}. \quad (19)$$

The expression increases with positive integer n , so its minimum occurs at $n = 1$:

$$\frac{\Delta x \Delta p}{\hbar} = \sqrt{\frac{\pi^2}{12} - \frac{1}{2}} = 0.567861808 \dots > \frac{1}{2}. \quad (20)$$

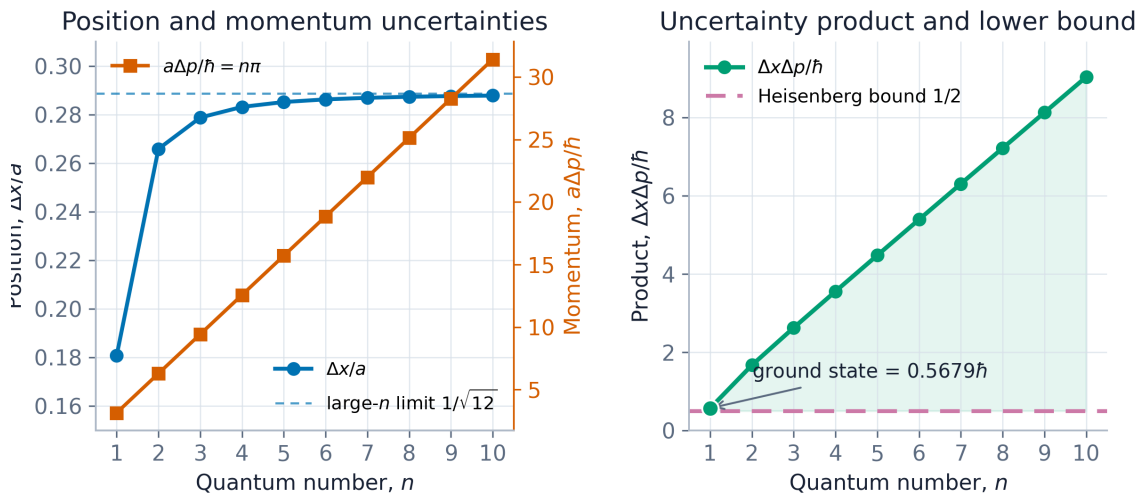
Therefore every stationary state in the box satisfies Heisenberg's relation

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (21)$$

The box ground state is not a minimum-uncertainty Gaussian state, so the inequality is strict.

Heisenberg uncertainty in the particle-in-a-box

$$\Delta x \Delta p / \hbar = \sqrt{n^2 \pi^2 / 12 - 1/2} \geq 1/2$$



The ground state gives the smallest product; momentum uncertainty then grows with quantum number.

Figure 2: Position and momentum uncertainties and their product for $n = 1, \dots, 10$. The dashed magenta line marks the Heisenberg bound.

6 Computational implementation and validation

The analytical model evaluates Eqs. (4)– (19) on a 2001-point position grid. A separate finite-difference model approximates the second derivative by

$$\left. \frac{d^2\psi}{dx^2} \right|_j \approx \frac{\psi_{j+1} - 2\psi_j + \psi_{j-1}}{(\Delta x)^2}, \quad (22)$$

creating a symmetric tridiagonal Hamiltonian. Its eigenvalues are calculated without inserting the analytical spectrum.

Table 1: Maximum relative energy error over the first ten numerical states.

Interior points	Maximum relative error
100	8.04×10^{-3}
200	2.03×10^{-3}
400	5.11×10^{-4}
800	1.28×10^{-4}
1600	3.21×10^{-5}

The observed convergence orders lie between 1.99896 and 1.99996, consistent with the expected second-order discretisation error. The finest-grid absolute overlap between every numerical eigenvector and its analytical counterpart is greater than 0.999999. The automated acceptance suite additionally checks normalisation, orthogonality, node counts, boundary values, energy scaling, expectation values and the uncertainty inequality.

7 Interpretation and limitations

The infinite well is an ideal model: real confining potentials are finite and may allow tunnelling. The calculation is one-dimensional, non-relativistic and contains one non-interacting particle with no external field or spin dynamics. These assumptions do not affect the internal verification of the specified Task 7 model, but they limit direct quantitative comparison with a particular laboratory system.

Reproducibility. The source code, CSV evidence, reference anchors, validation report, PNG/SVG figures and deterministic manifest are generated by a single command. The official brief is available from the British Physics Olympiad Computational Challenge page.